

FollowNet: A Comprehensive Benchmark for Car-Following Behavior Modeling

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Abstract—Car-following is a control process in which a following vehicle (FV) adjusts its acceleration to keep a safe distance from the lead vehicle (LV). Recently, there has been a booming of data-driven models that enable more accurate modeling of car-following through real-world driving datasets. Although there are several public datasets available, their formats are not always consistent, making it challenging to determine the state-of-the-art models and how well a new model performs compared to existing ones. In contrast, research fields such as image recognition and object detection have benchmark datasets like ImageNet, Microsoft COCO, and KITTI. To address this gap and promote the development of microscopic traffic flow modeling, we establish a public benchmark dataset for car-following behavior modeling. The benchmark consists of more than 80K car-following events extracted from five public driving datasets using the same criteria. These events cover diverse situations including different road types, various weather conditions, and mixed traffic flows with autonomous vehicles. Moreover, to give an overview of current progress in car-following modeling, we implemented and tested representative baseline models with the benchmark. Results show that the deep deterministic policy gradient (DDPG) based model performs competitively with a lower MSE for spacing compared to traditional intelligent driver model (IDM) and Gazis-Herman-Rothery (GHR) models, and a smaller collision rate compared to fully connected neural network (NN) and long short-term memory (LSTM) models in most datasets. The established benchmark will provide researchers with consistent data formats and metrics for cross-comparing different car-following models, promoting the development of more accurate models. We open-source our dataset and implementation code in <https://github.com/HKUST-DRIVE-AI-LAB/FollowNet>.

Index Terms—Car-following, Traffic Simulation, Traffic flow, Datasets, Benchmark.

I. INTRODUCTION

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CAR-FOLLOWING is the most fundamental and frequent driving behavior. It involves actions taken by a driver when following another vehicle ahead. Proper car-following behavior can lower crash risks and improve traffic flow stability [1]–[4]. The corresponding car-following model is a mathematical or computational representation of the behavior exhibited by drivers when following other vehicles on the road. It describes and predicts the dynamics of following vehicles (FV) and lead vehicles (LV) movements in traffic flow and is a cornerstone for microscopic traffic simulation [5], [6].

Over the past decade, there has been a boom of data-driven car-following models, primarily due to the availability of real-world driving data and advancements in machine learning. Representative data-driven car-following models include neural network based [7], [8], recurrent neural network (RNN) based [9], [10], and reinforcement learning (RL) based [11]–[14]. However, existing research has the following limitations:

- 1) **Lack of standardized data formats and evaluation criterion.** Although there are multiple public driving datasets available (e.g., NGSIM, HighD), there are no standardized car-following datasets and evaluation metrics, making it difficult to compare newly proposed car-following models with existing ones in terms of performance.
- 2) **Inadequate representation of car-following behavior in mixed traffic flows.** We will likely face a transitional phase of mixed traffic where autonomous and human-driven vehicles share the road. However, previous studies have mainly focused on modeling car-following behavior using limited datasets that do not include autonomous vehicles.

To further advance the field of microscopic traffic simulation modeling, it is imperative to establish a public car-following benchmark that can address the aforementioned issues and serve as a standard dataset for research in car-following. Such a benchmark would be beneficial for advancing the field of microscopic traffic flow modeling, similar to how standard datasets such as ImageNet [15], Microsoft COCO [16], and KITTI [17] have contributed to their respective fields. To achieve this, we have created the benchmark by extracting data from five public datasets based on the same criteria. Our benchmark includes five baseline car-following models, consisting of both traditional and data-driven models (Figure. 1). This paper presents a summary of the recent developments in this field, as well as evaluates the performance of mainstream models using our benchmark.

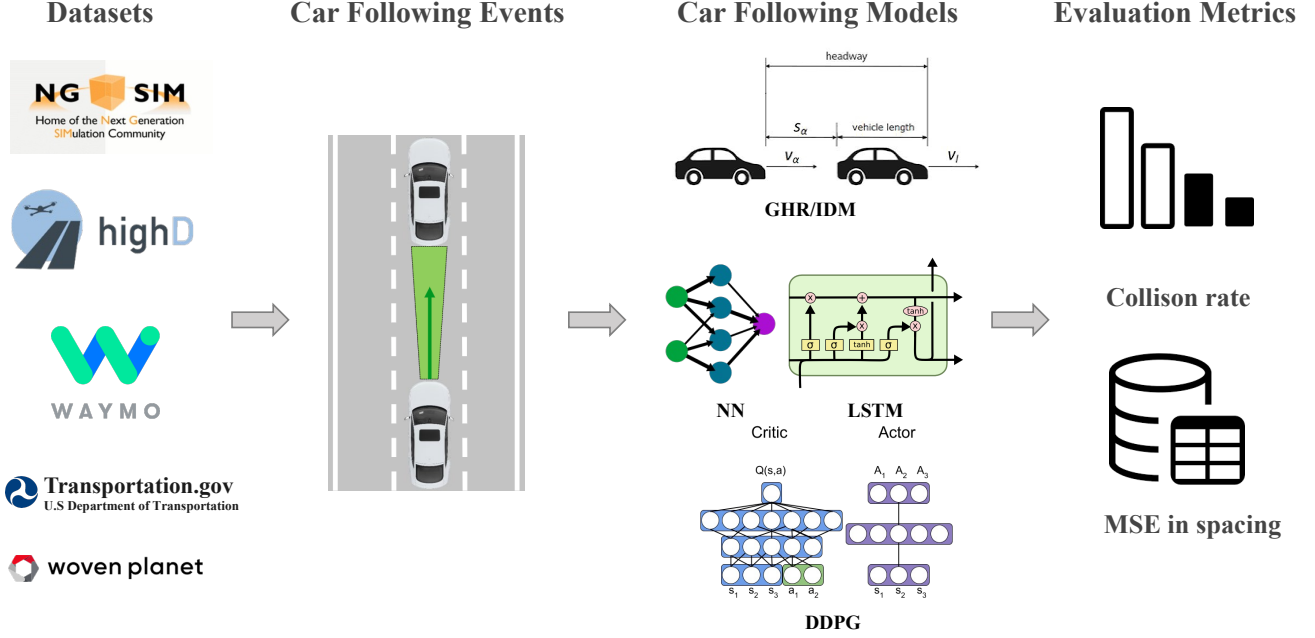


Fig. 1. A Roadmap of Car-following Benchmark: FollowNet

The contributions of this study include:

- Summarized existing car-following models systematically.
- Established the first benchmark of the car-following behavior to save the effort of data extraction and facilitate the development of car-following models.
- Offered diverse scenarios for testing car-following models, including mixed traffic and different road types, while evaluating these models using standardized metrics.
- Provided references for the community on car-following extraction, calibration, and baseline model implementations, through data and code open-sourcing.

The rest of the paper is organized as follows. Section II reviews car-following models. Section III analyzes the characteristics of car-following events in different datasets. Section IV introduces five baseline models. Section V evaluates the model performance with consistent metrics. Section VI discusses future directions of car-following research. Section VII concludes the paper.

II. RELATED WORK

A. Traditional Car-Following Models

Since the Pipes model was proposed in 1953 [18], researchers have studied car-following models for 70 years. In general, the existing traditional car-following models are divided into five main categories: stimulus-response model, safe distance model, psycho-physiological model, optimal velocity model, and desired goal model [19]–[21].

The stimulus-response model uses the relative distance or relative speed to determine the acceleration of the FV. The **Gazis-Herman-Rothery (GHR) model** [22], the first stimulus-response model, assumes that a vehicle's acceleration positively relates to the speed difference and that the FV's response

is expressed regarding its acceleration or deceleration behavior. The basic equation of this model can be expressed as:

$$a_n(t) = cv_n^m(t) \frac{\Delta v(t - T)}{\Delta x^l(t - T)} \quad (1)$$

where the n -th vehicle's acceleration at time t is represented by $a_n(t)$, Δv , and Δx are the speed difference and distance between the $(n - 1)$ th vehicle and the n th vehicle, respectively. m , c , and l are constants that need to be calculated, and T is the driver's reaction time. The GM car-following model [23] is another well-known stimulus-response model which used the difference between the LV and FV along with the velocity and space headway to determine the acceleration.

[24] proposed the first safety distance model. The underlying principle of the model differs from the stimulus-response model due to the LV's unpredictable motion. The model is also known as the conflict avoidance model since the FV keeps a minimum safe distance even though the LV implements sudden breaks. Its basic formula is:

$$\Delta x(t - T) = \alpha v_{n-1}^2(t - T) + \beta_l v_n^2(t) + \beta v_n(t) + b_0 \quad (2)$$

where Δx denotes the distance between the $(n - 1)$ th vehicle and the n -th vehicle, $v_n(t)$ is the speed of the n -th vehicle at time t , T is the driver's reaction time, α , β_l , β , b_0 are parameters to be calibrated. Similarly, **Gipps model** [5] utilizes the concept of the safe distance to facilitate car-following behavior. Specifically, the driver of the FV selects a certain speed and maintains a corresponding distance from the LV to prevent a collision.

The psycho-physiological model suggests that drivers adopt strategies based on the relative motion between the LV and FV, including changes in speed and distance differences, and only react when the threshold value is exceeded. This model was

first proposed in [25], and is used in VISSIM[®], a microscopic traffic simulation tool, through the Wiedemann model [26].

In [27], the optimal velocity model (OVM) was introduced for the first time, which can account for various traffic flow phenomena, such as free flow, congested traffic, the relationship between density and traffic flow, and stop-and-go traffic waves. However, the OVM model may produce unrealistic acceleration and deceleration processes. To address this limitation, the generalized force (GF) model [28] was introduced, which added the effect of negative speed difference to the optimal velocity model. However, both models neglect situations where the speed of the LV is much slower than that of the FV. The full velocity difference (FVD) model [29] considers the safety distance and offers a more precise acceleration function, which is given by:

$$a_n(t) = \alpha [V_n^*(\Delta X_n(t)) - V_n(t)] + \lambda (\Delta V_n(t))$$

$$\lambda = \begin{cases} \lambda_0 : & \Delta X_n(t) \leq s_c \\ 0 : & \Delta X_n(t) > s_c \end{cases} \quad (3)$$

The first term in the acceleration function is proportional to the difference between the optimal velocity $V_n^*(\Delta X_n(t))$ and the actual velocity $V_n(t)$, and the second term considers the velocity difference $\Delta V_n(t)$ as a linear stimulus. The sensitivity coefficients are denoted by α and λ , and s_c is a threshold value that distinguishes between following and free driving.

The desired goal model assumes that the driver has certain desired goals, such as desired following speed, desired headway, etc. **The Intelligent Driver Model (IDM)** [30] is the most frequently used driver-based desired goal model. It is also by far the most thorough and succinct theoretical car-following model that is free of accidents which assumes that each driver has a set of desired parameter values during a period of following behavior trying to maintain. The model expressions are:

$$a_n(t) = a_0 \left[1 - \left(\frac{v_n(t)}{\tilde{v}_n} \right)^\lambda - \left(\frac{\tilde{S}_n(t)}{s_n} \right)^2 \right] \quad (4)$$

$$\tilde{S}_n(t) = S_0 + v_n(t)\tilde{T} + \frac{v_n(t)\Delta v(t)}{2\sqrt{a_0 b}} \quad (5)$$

$$\Delta v(t) = v_n(t) - v_{n-1}(t) \quad (6)$$

where $a_n(t)$ and $v_n(t)$ represent the acceleration and the velocity of the FV at time t , respectively. And $S_n(t)$, $\Delta v(t)$ are the spacing and relative speed between the FV and the LV. The desired maximum acceleration, maximum deceleration, desired velocity, and desired time headway are represented by a_0 , b , $\tilde{v}$, and $\tilde{T}$, respectively. S_0 is the minimum safe headway and λ is a constant to be calibrated. Subsequently, the Intelligent Driver Model with Memory (IDMM) [31] is introduced which designs the IDM to incorporate memory effects and adapt driving behavior to the surrounding traffic.

Traditional car-following models have several disadvantages, which can be summarized into three main points:

- 1) **Simplistic and unrealistic assumptions.** Many traditional car-following models oversimplify driver behavior and do not reflect real-world conditions.

- 2) **Limited adaptability and data availability.** Traditional car-following models may lack the ability to adapt to dynamic traffic flow, and their effectiveness may be limited by the amount of data available.
- 3) **Limited applicability.** Traditional car-following models are often only applicable to a specific type of road and may not be able to capture the complexity of mixed-use road networks or account for emerging driving technologies beyond human-driver behavior.

More advanced car-following models have been developed that are able to get around some of these limitations as a result of recent advancements in technology as well as data collection techniques. However, in the study of traffic engineering and flow analysis, traditional ones are still very important and frequently employed.

B. Data-Driven Car-Following Models

Traditional models and data-driven models are the two main categories of car-following models, as shown in Figure. 2. Data-driven models make use of artificial intelligence techniques such as nonparametric regression, fully connected neural networks, recurrent neural networks, reinforcement learning, and other methods to predict drivers' behavior. These models learn relationships between different factors and the driver's behavior from the collected data.

A straightforward k-nearest neighbor (KNN)-based non-parametric car-following model [32] was presented in 2015 that forecasts the most likely driving behavior under the given circumstances. Similarly, Loess model [33] is also a non-parametric data-driven car-following model based on locally weighted regression.

Another common approach to predict the acceleration of an FV is to use a fully connected neural network (NN). A 4-layer neural network with two hidden layers [34] takes the gap distance, the relative velocity, desired velocity, and follower velocity as inputs to predict the FV's acceleration directly. In [35], the reactive agent-based car-following models were developed using artificial neural network (ANN) techniques, including backpropagation and fuzzy ARTMAP architectures. The results showed that the models' degrees of accuracy were best for the backpropagation and fuzzy ARTMAP architectures. Also, in [8], one hidden layer neural network can accurately predict acceleration and come up with the fuzzy rule-based neural network (FRNN), which was further improved by giving the neural network an instantaneous reaction time (RT) delay [36]. In [37], a Back-Propagation Neural Networks (BPNN) model combined with the Gipps model is proposed to avoid crashes.

Recurrent neural networks are another popular network structure that can model temporal problems such as vehicle following using historical information. Vanilla RNN [9] was utilized to simulate drivers' car-following behavior and demonstrated effective traffic oscillation prediction. Two other popular RNN variants, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) have also been used to model car-following behavior. [38] utilized LSTM to address issues of gradient vanishing and exploding during long sequence training, resulting in improved performance of the car-following